MIL-S-63383(AR)
AMENDMENT 3
22 November 1982
SUPERSEDING
AMENDMENT 2
31 March 1982

MILITARY SPECIFICATION

SAFE AND ARMING (NON-HAZARDOUS) ASSEMBLY
FOR MINES, M718/M741

This amendment forms a part of Military Specification MIL-S-63383(AR), dated 13 March 1980, and is approved for use by the U. S. Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

Delete:

"Safe and Arming (Clearing Charge Train) Assembly" and substitute, "Safe and Arming (Non-Hazardous) Assembly" wherever mentioned.

PAGE 1

2.1, Under SPECIFICATIONS, delete word "FEDERAL" and substitute "MILITARY".

Add: "MIL-S-48747 - Safe and Arming Device for Mines, M718/M741 System"

PAGE 2

3.3, Add to the end of the sentence:

"as evidenced by the presence of a length of solder in the dummy center lead and the absence of main charge leads and clearing charge propellant."

3.3, Delete from second sentence:

"as well as flexible cable assembly".

PAGE 3

3.4, Delete the second line and substitute:

"aft ejection charge assembly (without M5 propellant) shall all function high order when the detonators".

FSC 1345

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3.4.1, Delete paragraph in its entirety.

PAGE 4

Add: "-2" after 9277986 and 9277858 in drawing column.

Paragraph 4.3.1 Submission: Delete Cable and Detonator Assembly.

PAGE 5

TABLE I, First Article Inspection: Delete Cable and Detonator Assembly.

PAGE 6

Add: "(without M5 propellant)" after "Aft Ejection Charge Assembly" and "Chamber Assembly" in Examination or Test box.

Add: "-2" after drawing no. 9277858 and drawing no. 9277986 in Examination or Test box.

PAGE 7

4.4.1, Delete paragraph in its entirety and substitute the following new paragraph:

"4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, lots of safe and arming assemblies and sub-assemblies shall contain:

a. Parts bearing the same lot interfix number from one manufacturer and drawn from an accepted lot of representative parts used in the assembly of one lot of safe and arm assemblies, part number 9278015.

b. Explosive detonators, transfer leads, and aft-ejection charge assemblies (without M5 propellant) from the same lot interfix number from one manufacturer and drawn from an accepted lot of representative parts used in the assembly of one lot of safe and arm assemblies, part number 9278015.

Lot numbering shall comply with the lot numbering provisions of MIL-A-48078. Each lot of the Safe and Arming (Non-Hazardous Assemblies) shall be assigned a lot interfix number from within the 500 to 599 number group. All other lot number information shall be consistent with the lot number of safe and arming assembly lot, part number 9278015, from which the representative parts have been selected.

For example, if the Safe and Arming Assembly, part number 9278015, has a lot number of ABC80A001-001, the representative non-hazardous safe and arming assemblies, part number 9317620, would have a lot number of ABC80A500-001."

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4.4.2.2, Safe and Arming (Non-Hazardous) Subassembly

Critical Defect 2, revise to read: "Holder, spring, missing, improperly oriented, or loose".

Add Major Defect 107 as follows: Closure, Slider Oriented Properly, .40%, 3.2, Visual

Add the following critical defect:

Category	Exam or Test	No. of Sample Units	AQL	Reqm't Para.	Para. Ref.
"11.	Deformation of centrifugal lock hole(s) after spring holder staking	4.4.3.3	-	3.2	4.5.9"

Add "(5)" after the word "critical" in the category block.

Add to the notes:

"(5) paragraph 4.5.10 applies whenever critical defects are discovered."

4.4.2.3, Delete paragraph in its entirety.

4.4.2.4 - Add "(3)" after the word "critical" in the category block.

Add to the Notes block:

"(3) paragraph 4.5.10 applies whenever critical defects are discovered."

4.4.2.4: Critical defect 1 - In Examination or Test box - delete "(radiographic)" and substitute "(Visual and radiographic)". In Para Ref box - delete "4.5.4" and substitute "4.5.4/Visual".

4.4.2.4, Safe and Arming (Non-Hazardous) Assembly:

Delete Major Defect 104.

Add new Major Defect 104 as follows:

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"Detonator(s) not fully seated, 100%, 3.2, Visual/Manual."

Delete Major Defect 105.

Add new Major Defect 105 as follows:

"Solder improper, 100%, 3.2, Visual."

Add Major Defect 115 as follows:

Detonator placement incorrect, 0.40%, 3.2, Visual."

Add Major Defect 116 as follows:

"Height of solder joint(s), max., 0.40%, 3.2, Gage."

Add Major Defect 117 as follows:

"Solder workmanship under flexible cable improper, --, --, 4.4.3.4/4.5.11."

PAGE 13

4.4.2.5, Add "-2" after drawing number 9277986 in drawing number box.

4.4.2.5, Chamber Assembly Major Defect 101, revise to read:

"Aft ejection charge assembly missing or not fully seated in dual MDF chamber."

Delete Major Defect 104.

Add new Major Defect 104 as follows:

"Molded insert(s) missing or improperly seated, 0.40%, 3.2, Visual."

PAGE 14

4.4.2.6, Add "-2" after drawing number 9277858 in drawing number box.

Delete Major Defect 101 in its entirety.

PAGE 16

4.4.3.1, Delete paragraph in its entirety.

Add the following new paragraph:

"4.4.3.3 Centrifugal lock hole inspection - The inside diameter of the centrifugal lock holes in the S&A housing shall be inspected after staking of the centrifugal lock spring holders. The inspection shall be performed on three (3) centrifugal lock/housing

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subassemblies from each staking tool at the start of each shift and then on one (1) subassembly randomly selected from each group of 100 produced continuously from lock staking tool. The contractor shall not enter into production with any staking tool unless all three (3) samples pass the initial inspection. Both centrifugal lock holes on each sample shall be inspected. Inspection shall be performed in accordance with 4.5.9. The contractor may request a change in frequency of the inspection provided he presents substantive evidence to support the request."

4.4.3.4, Add the following paragraph as follows:

"4.4.3.4 Solder workmanship. The contractor shall assure that the solder process/procedure does not contaminate or create potential short circuit conditions on the bottom side of the flexible cable due to solder flow thru the holes. The contractor shall examine the interface between the detonators and the cable for flux, excessive solder or other unacceptable criteria specified in the solder standard listed on the drawing. The examination shall be performed, after removal from the housing, on five (5) detonator/cable assemblies prior to the start of each shift, after every two (2) hours, and at the end of the shift until three (3) consecutive shifts have been accepted. Thereafter, the examination may be performed on two (2) assemblies prior to the start of each shift and one sample after four (4) hours of production and at the end of the shift. Failure of a sample at any time will cause the examination to revert to the initial conditions until three (3) consecutive shifts have been accepted."

* 4.5.4, Delete paragraph 4.5.4 in its entirety and substitute new paragraph 4.5.4 as follows:

"4.5.4 Radiographic examination. The safe and arming assemblies radiographic plates shall be of a quality such that the Technical Agency of 6.5 or their appointed representatives shall be able to view all of the requirements to the Agency's satisfaction. The radiographic plates shall clearly define all explosive components; positively show that the assembly is in the full non-armed condition; the flexible cable shall not be pinched, twisted or damaged; the MDF assembly shall not show evidence of any damage and all components shall be in the non-armed position and the S&A explosive loading complies with the requirement of 3.3. Any assembly failing to comply with all of the specified requirements shall be classed defective and removed from the lot. The vendor will have the option of substituting a paper base sensitive emulsion for the film base (Ref. MIL-STD-453) provided that this material produces the required definition specified above.
(Non-destructive test)."

4.5.5.1: Delete paragraph 4.4.5.1 in its entirety and substitute new paragraph 4.4.5.1 as follows:

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"4.5.5.1 Detonator continuity resistance. The measurement of the detonator resistance (value specified on Dwg 9317620) shall be obtained on the final safe and arming assembly, (including staked shield assembly) utilizing an approved milliohmeter whose excitation current does not exceed the no fire condition of the detonators. The resistance measurement shall be obtained across the eyelets on the flexible cable (Dwg. 9277774). Any assembly failing to comply with the specified resistance requirement shall be classed defective. The exposed conductor (shorting loop) in the flexible cable (Dwg. 9277774) must be cut to perform this test and soldered to reconnect the above conductor after completion of test. (Non-destructive test)."

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4.5.6.2, Delete paragraph in its entirety.

PAGE 21

Delete page 21 in its entirety.

PAGE 22

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PAGE 23

Delete page 23 in its entirety.

PAGE 24

Delete page 24 in its entirety.

PAGE 25

Delete page 25 in its entirety.

PAGE 26

Delete page 26 in its entirety.

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Add the following new paragraphs:

"4.5.9 Centrifugal lock hole inspection. Both centrifugal lock spring holders shall be removed using approved tooling and procedures. The housing shall be rotated and the centrifugal locks allowed to fall from the housing.

A .2495 - .005 inch pin gage shall fall freely (without applying pressure) to the bottom of the hole (shoulder). If one or both locks fail to fall out under their own weight or the pin gage fails to insert freely in one or both holes, the unit shall be classed defective and that quantity of subassemblies from which the sample was drawn shall be rejected (Destructive test).

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4.5.10 Critical defect reporting. The contractor shall proceed as follows whenever a critical defect is discovered during the examinations or tests described in this specification:

a. Immediately notify the cognizant technical agency.

b. Tentatively reject and quarantine all previously accepted S&A subassemblies in process.

c. Determine root cause of defect and proceed as directed by the contracting office."

4.5.11, Add paragraph as follows:

"4.5.11 Solder workmanship. Examination for solder workmanship on the bottom side of the cable shall be performed on detonators and flexible cables and grounding clips soldered in accordance with the procedures normally followed during production with the detonators and grounding clips contained in re-usable housing (P/N 9278008) that contain no components and that have been modified by enlarging the detonator holes to facilitate removal of the detonators. The contractor shall recommend the extent of the housing modification and request approval in accordance with para. 6.3. Any unit failing to comply with the soldering requirements at the start of the shift shall result in the contractor not proceeding with production until corrective action is taken and a subsequent sample passes. Any sample failing during the subsequent examinations or at the end of the shift shall result in all production following the last successful examination being rejected. (Destructive examination)."

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Add to paragraph 6.2:

"A quantity of Safe and Arming (Non-Hazardous) Assemblies, P/N 9317620 shall be fabricated with each lot of Safe and Arming Assemblies, P/N 9278015 manufactured. Quantity and requirement for delivery shall be as specified in the contract."

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire contect irrespective of the marginal notations and relationship to the last previous amendment.

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